<u>The hands-on part of this session combines:</u>

(1) Preparation of the project data

(2) Practicing handling and processing of tabular data using the `pandas` library

Step1: Loading the dataset

<u>Preferably:</u>

Load the dataset directly from the scikit-learn repository

```
import pandas as pd
from sklearn.datasets import load_diabetes

diabetes = load_diabetes(scaled=False)
df_diabetes_raw = pd.DataFrame(data=diabetes.data, columns=diabetes.feature_names)
df_diabetes_raw['target'] = diabetes.target
```

Note: This should be the way in which the data is imported in your Jupyter Notebook for the final project report.

<u>Alternatively, for the sake of the exercise, as a temporary solution</u>:

Load the data from the file deposited in the canvas board, as follows:

```
df_diabetes_raw = pd.read_csv("Project_data_sklearn_diabetes.csv", index_col = 0)
```

<u>In both ways, the following data frame should be obtained at the end (verify this):</u>

```
df_diabetes_raw
```

	age	sex	bmi	bp	s1	s2	s3	s4	s5	s6	target
0	59.0	2.0	32.1	101.00	157.0	93.2	38.0	4.00	4.8598	87.0	151.0
1	48.0	1.0	21.6	87.00	183.0	103.2	70.0	3.00	3.8918	69.0	75.0
2	72.0	2.0	30.5	93.00	156.0	93.6	41.0	4.00	4.6728	85.0	141.0
3	24.0	1.0	25.3	84.00	198.0	131.4	40.0	5.00	4.8903	89.0	206.0
4	50.0	1.0	23.0	101.00	192.0	125.4	52.0	4.00	4.2905	80.0	135.0
...	...	...	...	...	...	...	...	...	...	...	...
437	60.0	2.0	28.2	112.00	185.0	113.8	42.0	4.00	4.9836	93.0	178.0
438	47.0	2.0	24.9	75.00	225.0	166.0	42.0	5.00	4.4427	102.0	104.0
439	60.0	2.0	24.9	99.67	162.0	106.6	43.0	3.77	4.1271	95.0	132.0
440	36.0	1.0	30.0	95.00	201.0	125.2	42.0	4.79	5.1299	85.0	220.0
441	36.0	1.0	19.6	71.00	250.0	133.2	97.0	3.00	4.5951	92.0	57.0

442 rows × 11 columns

Step2: Renaming the columns

Rename the columns of `df_diabetes_raw` as follow (original name ---> new name):

age ---> Age
sex ---> Gender
bmi ---> BMI
bp ---> BP
s1 ---> TC
s2 ---> LDL
s3 ---> HDL
s4 ---> TCH
s5 ---> LTG
s6 ---> Glu
target ---> Progression

Hint: use the `.rename()` method of DataFrame

At the end of this stage you should obtain for `df_diabetes_raw`:

```
df_diabetes_raw
```

	Age	Gender	BMI	BP	TC	LDL	HDL	TCH	LTG	Glu	Progression
0	59.0	2.0	32.1	101.00	157.0	93.2	38.0	4.00	4.8598	87.0	151.0
1	48.0	1.0	21.6	87.00	183.0	103.2	70.0	3.00	3.8918	69.0	75.0
2	72.0	2.0	30.5	93.00	156.0	93.6	41.0	4.00	4.6728	85.0	141.0
3	24.0	1.0	25.3	84.00	198.0	131.4	40.0	5.00	4.8903	89.0	206.0
4	50.0	1.0	23.0	101.00	192.0	125.4	52.0	4.00	4.2905	80.0	135.0
...	...	...	...	...	...	...	...	...	...	...	...
437	60.0	2.0	28.2	112.00	185.0	113.8	42.0	4.00	4.9836	93.0	178.0
438	47.0	2.0	24.9	75.00	225.0	166.0	42.0	5.00	4.4427	102.0	104.0
439	60.0	2.0	24.9	99.67	162.0	106.6	43.0	3.77	4.1271	95.0	132.0
440	36.0	1.0	30.0	95.00	201.0	125.2	42.0	4.79	5.1299	85.0	220.0
441	36.0	1.0	19.6	71.00	250.0	133.2	97.0	3.00	4.5951	92.0	57.0

442 rows × 11 columns

Step3: Renaming the index

Rename the index of `df_diabetes_raw` index to obtain the following:

	Age	Gender	BMI	BP	TC	LDL	HDL	TCH	LTG	Glu	Progression
Patient_ID_000	59.0	2.0	32.1	101.00	157.0	93.2	38.0	4.00	4.8598	87.0	151.0
Patient_ID_001	48.0	1.0	21.6	87.00	183.0	103.2	70.0	3.00	3.8918	69.0	75.0
Patient_ID_002	72.0	2.0	30.5	93.00	156.0	93.6	41.0	4.00	4.6728	85.0	141.0
Patient_ID_003	24.0	1.0	25.3	84.00	198.0	131.4	40.0	5.00	4.8903	89.0	206.0
Patient_ID_004	50.0	1.0	23.0	101.00	192.0	125.4	52.0	4.00	4.2905	80.0	135.0
...	...	...	...	...	...	...	...	...	...	...	...
Patient_ID_437	60.0	2.0	28.2	112.00	185.0	113.8	42.0	4.00	4.9836	93.0	178.0
Patient_ID_438	47.0	2.0	24.9	75.00	225.0	166.0	42.0	5.00	4.4427	102.0	104.0
Patient_ID_439	60.0	2.0	24.9	99.67	162.0	106.6	43.0	3.77	4.1271	95.0	132.0
Patient_ID_440	36.0	1.0	30.0	95.00	201.0	125.2	42.0	4.79	5.1299	85.0	220.0
Patient_ID_441	36.0	1.0	19.6	71.00	250.0	133.2	97.0	3.00	4.5951	92.0	57.0

442 rows × 11 columns

Hint 1: to rename the index, use `df_diabetes_raw.index =` (and then the appropriate term)

Hint 2:

```
f"{1:03}"
'001'
```

```
f"{16:03}"
'016'
```

Hint 3:

```
[f"a_{x}" for x in [1, 2, 3, 4, 5]]
['a_1', 'a_2', 'a_3', 'a_4', 'a_5']
```

Step 4: Re-encoding the values in the Gender column

Change the values in the Gender column as follows (original value ---> renamed value):

1.0 ---> “Female”
2.0 --->. “Male”

Hint: You need first to change the data type of the Gender column, using `.astype()`

```
df_diabetes_raw["Gender"] = df_diabetes_raw["Gender"].astype("string")
```

At the end of this step 4 you should obtain:

	Age	Gender	BMI	BP	TC	LDL	HDL	TCH	LTG	Glu	Progression
Patient_ID_000	59.0	Male	32.1	101.00	157.0	93.2	38.0	4.00	4.8598	87.0	151.0
Patient_ID_001	48.0	Female	21.6	87.00	183.0	103.2	70.0	3.00	3.8918	69.0	75.0
Patient_ID_002	72.0	Male	30.5	93.00	156.0	93.6	41.0	4.00	4.6728	85.0	141.0
Patient_ID_003	24.0	Female	25.3	84.00	198.0	131.4	40.0	5.00	4.8903	89.0	206.0
Patient_ID_004	50.0	Female	23.0	101.00	192.0	125.4	52.0	4.00	4.2905	80.0	135.0
...	...	...	...	...	...	...	...	...	...	...	...
Patient_ID_437	60.0	Male	28.2	112.00	185.0	113.8	42.0	4.00	4.9836	93.0	178.0
Patient_ID_438	47.0	Male	24.9	75.00	225.0	166.0	42.0	5.00	4.4427	102.0	104.0
Patient_ID_439	60.0	Male	24.9	99.67	162.0	106.6	43.0	3.77	4.1271	95.0	132.0
Patient_ID_440	36.0	Female	30.0	95.00	201.0	125.2	42.0	4.79	5.1299	85.0	220.0
Patient_ID_441	36.0	Female	19.6	71.00	250.0	133.2	97.0	3.00	4.5951	92.0	57.0

442 rows × 11 columns

Step 5: Separate the `df_diabetes_raw` data frame to two data frames

Divide the df_diabetes_raw data frame to two data frames:

- `df_diabetes_metadata` containing the "Age", "Gender" features
- `df_diabetes_features` containing the other features ('BMI', 'BP', 'TC', 'LDL', 'HDL', 'TCH', 'LTG', 'Glu','Progression')

Both data frames should have the same index, as of the index of df_diabetes_raw (and in the same order)

At the end you should obtain:

```
df_diabetes_metadata
```

	Age	Gender
Patient_ID_000	59.0	Male
Patient_ID_001	48.0	Female
Patient_ID_002	72.0	Male
Patient_ID_003	24.0	Female
Patient_ID_004	50.0	Female
...	...	...
Patient_ID_437	60.0	Male
Patient_ID_438	47.0	Male
Patient_ID_439	60.0	Male
Patient_ID_440	36.0	Female
Patient_ID_441	36.0	Female

442 rows × 2 columns

```
df_diabetes_features
```

	BMI	BP	TC	LDL	HDL	TCH	LTG	Glu	Progression
Patient_ID_000	32.1	101.00	157.0	93.2	38.0	4.00	4.8598	87.0	151.0
Patient_ID_001	21.6	87.00	183.0	103.2	70.0	3.00	3.8918	69.0	75.0
Patient_ID_002	30.5	93.00	156.0	93.6	41.0	4.00	4.6728	85.0	141.0
Patient_ID_003	25.3	84.00	198.0	131.4	40.0	5.00	4.8903	89.0	206.0
Patient_ID_004	23.0	101.00	192.0	125.4	52.0	4.00	4.2905	80.0	135.0
...	...	...	...	...	...	...	...	...	...
Patient_ID_437	28.2	112.00	185.0	113.8	42.0	4.00	4.9836	93.0	178.0
Patient_ID_438	24.9	75.00	225.0	166.0	42.0	5.00	4.4427	102.0	104.0
Patient_ID_439	24.9	99.67	162.0	106.6	43.0	3.77	4.1271	95.0	132.0
Patient_ID_440	30.0	95.00	201.0	125.2	42.0	4.79	5.1299	85.0	220.0
Patient_ID_441	19.6	71.00	250.0	133.2	97.0	3.00	4.5951	92.0	57.0

442 rows × 9 columns

Step 6: Save the two data frames as Excel files

Save `df_diabetes_metadata` and `df_diabetes_features` as Excel files named `Table_diabetes_metadata.xlsx` and `df_diabetes_features.xlsx`, respectively.